

A Mathlib-Idiomatic Formalization of the Fricke Involution and its Operator on Modular Forms

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Status. This is a *preprint*. It has **not been peer reviewed**. The Lean development it describes compiles and is axiom-audited — those claims are machine-checked and independently reproducible — but the *exposition* has had no external referee. This version incorporates revisions from two internal adversarial passes: one against an earlier claim of mathematical priority (found false and retracted, §6) and one against an attempted T-duality bridge theorem (found unproved and reported as a negative result, §9); neither constitutes venue peer review. Read the scope before citing the result.

Abstract

We formalize, in Lean 4 on top of Mathlib, the Fricke involution $W_N = \begin{pmatrix} 0 & -1 \\ N & 0 \end{pmatrix}$ on $\Gamma_0(N)$ and the operator it induces on modular forms, as a small reusable API rather than as proof-internal infrastructure. We claim no priority: Fricke and Atkin–Lehner material already exists in Lean 4 in the Fermat’s Last Theorem artifact [10] (§6), which covers strictly more. What we offer is a packaging aimed at upstreaming — Mathlib itself contains no Fricke or Atkin–Lehner operator. Mathlib already provides the upper half-plane, the Möbius action of $GL_2(\mathbb{R})$, the weight- k slash action carrying the determinant factor, the congruence subgroups $\Gamma_0(N)$, and the bundled types `SlashInvariantForm` and `ModularForm`; it contains no Fricke or Atkin–Lehner operator. We supply that missing piece in three layers: (i) the matrix level — $W_N \in GL_2^+(\mathbb{R})$ with $\det W_N = N$, the involution identity $W_N^2 = -N \cdot I$, the explicit conjugation $W_N \gamma W_N^{-1} = \begin{pmatrix} d & -c/N \\ -Nb & a \end{pmatrix}$, and the theorem that W_N normalizes $\Gamma_0(N)$; (ii) the slash level — $f \mapsto f|_k W_N$ preserves $\Gamma_0(N)$ -slash-invariance; and (iii) the bundled level — the operator is well defined on `ModularForm`, its holomorphy and boundedness at the cusps transported along W_N . The development is 27 declarations in 291 lines; all 21 theorems compile with no `sorry` and depend on exactly Lean’s three standard axioms `propext`, `Classical.choice` and `Quot.sound`, pinned by 16 build-failing `#guard_msgs` checks so that a drifted footprint fails compilation. We make no claim of new mathematics: the results are classical, due to Fricke and to Atkin–Lehner [1]. The contribution is the mechanization, and its intended use is as infrastructure for formalizing eta-quotient identities, where Fricke eigenvalues are the organizing invariant.

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1 Introduction

The Fricke involution is elementary to state and ubiquitous in the theory of modular forms on $\Gamma_0(N)$. For $N \geq 1$ set

$$W_N = \begin{pmatrix} 0 & -1 \\ N & 0 \end{pmatrix}, \quad \det W_N = N > 0.$$

Acting on the upper half-plane by $\tau \mapsto -1/(N\tau)$, it exchanges the cusps 0 and ∞ of $\Gamma_0(N)$, normalizes $\Gamma_0(N)$, and induces an involution on $M_k(\Gamma_0(N))$ whose ± 1 eigenspaces organize much of the arithmetic of the space — most visibly for eta-quotients, where multiplicativity is read off from Fricke and Atkin–Lehner eigenvalues [7, 6].

This note is written in the mode that recent practice makes standard: a formal artifact shipped alongside the human exposition. Large-scale autoformalization has crossed a threshold — a complete machine-checked proof of Fermat’s Last Theorem was produced in eleven days by AI agents coordinating through a shared statement graph [10, 11] — and the surrounding methodological questions are now mainstream mathematics [12, 3]. We work in Lean 4 [15] against Mathlib [2], and we adopt two specific practices from the FLT artifact: the axiom audit as a *build target*, and an explicit statement-dependency graph (§5).

What Mathlib already provides

We are explicit about prior art, because it determines what is and is not new here. As of commit 905b9581 (2026-07-28), Mathlib [2] contains, in forms strictly more general than we would need to restate:

- `UpperHalfPlane` and the Möbius action of `GL (Fin 2) ℝ`, together with `denom_ne_zero_of_im` and `normSq_denom_pos`;
- `im_smul_eq_div_normSq`, which is exactly the classical identity $\text{Im}(M \cdot \tau) = |\det M| \text{Im}(\tau) / |c\tau + d|^2$, stated for arbitrary $M \in GL_2(\mathbb{R})$ via $|\det M|$ rather than only for $\det M > 0$;
- `Matrix.GLPoS`, the subgroup of positive determinant;
- `CongruenceSubgroup.Gamma0 N`;
- the weight- k slash action `SlashAction ℤ (GL (Fin 2) ℝ) (ℍ → ℂ)`, whose definition *already carries* the determinant factor, $f|_k g = \sigma(g) f(g \cdot \tau) |\det g|^{k-1} \text{denom}(g, \tau)^{-k}$;
- the bundled `ModularForm Γ k` for $\Gamma : \text{Subgroup } (GL (Fin 2) \mathbb{R})$, with holomorphy and boundedness at cusps.

None of that is reproved here; we import it. A reader looking for a formalization of the $GL_2^+(\mathbb{R})$ -action on $\mathbb{H}$ should use Mathlib’s, not ours.

The gap, and what we contribute

A repository-wide search of Mathlib at that commit for `Fricke` or `AtkinLehner` returns *zero* occurrences: the machinery to host these operators exists in Mathlib, but the operators do not. That is the gap we address — with the important qualification, developed in §6, that Fricke material *does* already exist in Lean 4 outside Mathlib. Our contributions are therefore stated as an API, not as firsts:

(C1) W_N as an element of `GL (Fin 2) ℝ` with $\det W_N = N$, and $W_N \in GL_2^+(\mathbb{R})$ (Theorem 3.1);

- (C2) $W_N^2 = -N \cdot I$ (Theorem 3.2), which since scalar matrices act trivially on $\mathbb{H}$ is the involution property;
- (C3) the explicit conjugation formula (Theorem 3.3);
- (C4) W_N conjugates $\Gamma_0(N)$ into itself inside $GL_2(\mathbb{R})$ (Theorem 3.4); together with $W_N^2 = -N \cdot I$ central this gives the two-sided normalizer condition, though we formalize the containment;
- (C5) slashing by W_N preserves $\Gamma_0(N)$ -slash-invariance, giving the operator on `SlashInvariantForm` (Theorem 4.1);
- (C6) holomorphy and boundedness at the cusps transport along W_N , giving the operator on the bundled `ModularForm` (Theorem 4.2);
- (C7) the composite $f \mapsto (f|_k W_N)|_k W_N$ is multiplication by the explicit scalar $(-1)^k N^{k-2}$ (Theorem 4.3), which identifies the normalization making $|_k W_N$ an involution.

(C4) is the hinge: it is what makes $f \mapsto f|_k W_N$ well defined at all. (C5) and (C6) carry it up to Mathlib’s bundled types, so that the Fricke operator can be composed with the rest of the library rather than living beside it.

2 Mathematical background

Let $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL_2(\mathbb{Z})$ with $\gamma \in \Gamma_0(N)$, i.e. $N \mid c$; write $c = Nc'$. A direct computation gives

$$W_N \gamma W_N^{-1} = \begin{pmatrix} d & -c' \\ -Nb & a \end{pmatrix}, \quad (1)$$

whose entries are integers precisely because $N \mid c$, whose determinant is $da - (-c')(-Nb) = ad - bc = 1$, and whose lower-left entry $-Nb$ is divisible by N . Hence the conjugate again lies in $\Gamma_0(N)$, and W_N normalizes $\Gamma_0(N)$.

Rather than manipulate W_N^{-1} in the formalization, we prove the equivalent *intertwining* identity

$$W_N \cdot \gamma = \delta \cdot W_N, \quad \delta := \begin{pmatrix} d & -c' \\ -Nb & a \end{pmatrix}, \quad (2)$$

which avoids inverses entirely and from which (1) follows by one right multiplication. Both sides of (2) equal $\begin{pmatrix} -c & -d \\ Na & Nb \end{pmatrix}$, the two agreeing exactly on the relation $c = Nc'$.

3 The Lean development

The development is four modules. We reproduce the load-bearing declarations verbatim from the compiled source.

```
def frickeMatrix (N : ℕ) : Matrix (Fin 2) (Fin 2) ℝ := !![0, -1; (N : ℝ), 0]

@[simp] theorem frickeMatrix_det (N : ℕ) : (frickeMatrix N).det = (N : ℝ) := by
  simp [frickeMatrix, Matrix.det_fin_two_of]

noncomputable def frickeW {N : ℕ} (hN : 0 < N) : GL (Fin 2) ℝ :=
  Matrix.GeneralLinearGroup.mkOfDetNeZero (frickeMatrix N) (frickeMatrix_det_ne_zero hN)
```

Listing 1: The Fricke matrix and its determinant.

Theorem 3.1 (W_N has positive determinant). *frickeW_mem_GLPos : frickeW hN ∈ GLPos (Fin 2) ℝ.*

Theorem 3.2 (Involution). $W_N^2 = -N \cdot I$: *frickeW_sq_coe : ((frickeW hN * frickeW hN : GL (Fin 2) ℝ) : Matrix _ _ ℝ) = -(N : ℝ)) • 1.*

Since scalar matrices act trivially on $\mathbb{H}$, Theorem 3.2 says precisely that W_N induces an involution of the upper half-plane.

The arithmetic input is the divisibility extracted from membership in $\Gamma_0(N)$, which Mathlib states modulo N :

```
theorem gamma0_dvd_lower_left {N : ℕ} {γ : SL(2, ℤ)} (hγ : γ ∈ Gamma0 N) :
  (N : ℤ) ∣ γ.1 1 0 := by
  rw [Gamma0_mem] at hγ
  exact (ZMod.intCast_zmod_eq_zero_iff_dvd _ _).mp hγ
```

Listing 2: Divisibility of the lower-left entry.

The conjugate δ of (1) is then definable as a genuine $SL_2(\mathbb{Z})$ element, its determinant condition discharged by `linear_combination` against $\det \gamma = 1$:

```
def frickeConj {N : ℕ} (γ : SL(2, ℤ)) (c' : ℤ) (hc : γ.1 1 0 = (N : ℤ) * c') : SL(2, ℤ) :=
  (!![[γ.1 1 1, -c'; -(N : ℤ) * γ.1 0 1, γ.1 0 0], by
    have hdet := γ.2
    rw [Matrix.det_fin_two] at hdet; rw [hc] at hdet
    rw [Matrix.det_fin_two_of]; linear_combination hdet)
```

Listing 3: The Fricke conjugate, with its determinant proof.

Theorem 3.3 (Conjugation). *frickeW_conj_eq : for $\gamma \in SL_2(\mathbb{Z})$ with $c = Nc'$,*

$$W_N \cdot \text{mapGL}_{\mathbb{R}}(\gamma) \cdot W_N^{-1} = \text{mapGL}_{\mathbb{R}}(\text{frickeConj } \gamma c').$$

The proof rewrites by the intertwining identity (2) and cancels $W_N W_N^{-1}$.

Theorem 3.4 (Main result: normalization). *frickeW_normalizes_Gamma0 :*

$$\forall g \in \text{Gamma0}(N). \text{map}(\text{mapGL}_{\mathbb{R}}), \quad W_N g W_N^{-1} \in \text{Gamma0}(N). \text{map}(\text{mapGL}_{\mathbb{R}}).$$

Proof (as formalized). Destructure $g = \text{mapGL}_{\mathbb{R}}(\gamma)$ with $\gamma \in \Gamma_0(N)$; obtain c' with $c = Nc'$ from `gamma0_dvd_lower_left`; rewrite by Theorem 3.3; conclude by `Subgroup.mem_map_of_mem` applied to `frickeConj_mem_Gamma0`, whose content is that the lower-left entry $-Nb$ vanishes modulo N . $\square$

4 The operator on modular forms

Normalization is a statement about groups; being an *operator on modular forms* requires carrying the two analytic side conditions of `ModularForm` along W_N . We do this in two steps, working with the image $\Gamma_0(N)_{\mathbb{R}} := \Gamma_0(N). \text{map}(\text{mapGL}_{\mathbb{R}})$ inside $GL_2(\mathbb{R})$, which is the subgroup Mathlib’s bundled types are stated over.

Theorem 4.1 (Slash-invariance). *slash_frickeW_invariant : for f a `SlashInvariantForm` of weight k for $\Gamma_0(N)_{\mathbb{R}}$ and any g in that subgroup, $(f|_k W_N)|_k g = f|_k W_N$.*

Proof (as formalized). Write $g = \text{mapGL}_{\mathbb{R}}(\gamma)$ with $\gamma \in \Gamma_0(N)$ and extract $c = Nc'$. Then, using the slash cocycle in both directions and the intertwining identity (2),

$$(f|_k W_N)|_k \gamma = f|_k(W_N \gamma) = f|_k(\delta W_N) = (f|_k \delta)|_k W_N = f|_k W_N,$$

the last step by slash-invariance of f under $\delta \in \Gamma_0(N)$ (Theorem 3.3). This packages as `frickeSlashOperator`. $\square$

Theorem 4.2 (Operator on `ModularForm`). *`frickeModularOperator` : $f \mapsto f|_k W_N$ is a map `ModularForm` $\Gamma_0(N)_{\mathbb{R}} k \rightarrow \text{ModularForm } \Gamma_0(N)_{\mathbb{R}} k$.*

Proof (as formalized). Holomorphy is immediate from Mathlib’s `MDifferentiable.slash`: slashing by any fixed element of $GL_2(\mathbb{R})$ preserves `MDiff`. For boundedness, Mathlib’s `OnePoint.IsBoundedAt.smul_iff` reduces *$f|_k W_N$ is bounded at c* to *f is bounded at $W_N \cdot c$* , so it suffices that W_N carries cusps of $\Gamma_0(N)$ to cusps of $\Gamma_0(N)$. That is `isCusp_frickeW_smul`, obtained from Mathlib’s `isCusp.smul` together with the subgroup form of Theorem 3.4 (`frickeW_conjAct_le`) and the monotonicity of `isCusp` in its subgroup argument. $\square$

Concretely W_N exchanges the cusps 0 and ∞ of $\Gamma_0(N)$; what is proved here is the weaker statement, sufficient for the transport, that the cusp set is preserved.

Theorem 4.3 (The composite is a scalar). *`frickeW_sq_slash` : for every $f : \mathbb{H} \rightarrow \mathbb{C}$ and every weight k ,*

$$(f|_k W_N)|_k W_N = (N^2)^{k-1}(-N)^{-k} f = (-1)^k N^{k-2} f.$$

Proof (as formalized). By the cocycle law the left side is $f|_k(W_N^2)$, and $W_N^2 = -N \cdot I$ is scalar (Theorem 3.2). For that matrix the three ingredients of Mathlib’s slash are computed directly: $\det(W_N^2) = N^2$, $\text{denom}(W_N^2, \tau) = -N$, and W_N^2 acts trivially on $\mathbb{H}$. The conjugation factor σ needs no determinant case split: $\sigma(gg') = \sigma g \circ \sigma g'$ and $\sigma g \circ \sigma g = \text{id}$ give $\sigma(W_N^2) = \text{id}$ at once. $\square$

For even k it follows that $f \mapsto N^{1-k/2}(f|_k W_N)$ squares to the identity — the Fricke involution proper. For odd k the question is vacuous, since $-I \in \Gamma_0(N)$ forces $f \equiv 0$ (Mathlib’s `ModularForm.eq_zero_of_neg_one_mem`).

5 Verification status

Every declaration in the module compiles. There are no `sorry` placeholders and no project-local axioms. The axiom footprint of each theorem, obtained by `#print axioms`, is exactly

[`propext`, `Classical.choice`, `Quot.sound`],

the three standard axioms of Lean’s logic. Following the Fermat’s Last Theorem artifact [10], the footprint is not merely reported but *enforced*: a `FinalCheck` module places a `#guard_msgs in #print axioms` guard on each of the sixteen headline theorems inside the default build target, so any widening of the axiom set is a compile error, and a negative control (a guard asserting a deliberately wrong footprint) is kept outside the build and verified to fail. The development is tracked as a statement-dependency graph (`dag/theorems.jsonl`, with a human-readable map in `dag/PROOF-PATH.md`); the machine checks that every node marked `proved` names an existing declaration and depends only on `proved` nodes. We claim nothing beyond what the kernel checks: in the terminology we adopt throughout the SocrateAI programme, (C1)–(C7) are *formal* results; the surrounding exposition is not.

Artifact. Archived at Zenodo, [doi:10.5281/zenodo.22542571](https://doi.org/10.5281/zenodo.22542571), which is the citable version of record for this note and its sources. Repository <https://github.com/xaviecallens/SocrateAI-Lean-Lib> (tag `fricke-v1`); module `Lean/SocrateAI/ModularForms/{FrickeInvolution, FrickeSlash, FrickeModular, FrickeComposite}.lean`, with the axiom guards in `Lean/SocrateAI/FinalCheck.lean`; a mirror with a dataset card is at <https://huggingface.co/datasets/callensxavier/socrateai-fricke-involution>. Toolchain `leanprover/lean4:v4.32.2`; Mathlib pinned at commit `905b9581` (2026-07-28); `lake build SocrateAI` completes in 3053 jobs (including the axiom-guard module). The build configuration is distributed and portable: `lakefile.lean` requires Mathlib from git at the pinned revision above, `lean-toolchain` matches it, and no tracked file contains a machine-specific path. `git clone, lake exe cache get, lake build SocrateAI` reproduces the artifact; see `BUILDING.md`. Earlier versions of this note stated that the configuration was not distributed. That was true, and understated: the published `lakefile.lean` declared no Mathlib dependency at all and its `lean-toolchain` named a different compiler version, so a clean clone could not build. Both are fixed. The four modules total 27 declarations in 291 lines and import only `Mathlib.NumberTheory.ModularForms.SlashActions`, `Mathlib.NumberTheory.ModularForms.CongruenceSubgroups`, `Mathlib.NumberTheory.ModularForms.SlashInvariantForms` and `Mathlib.NumberTheory.ModularForms.Basic`. A statement-dependency graph with a natural-language description per node, and a validator enforcing that every node marked `proved` names an existing declaration and depends only on proved nodes, is in `dag/`.

6 Related work

Prior Lean 4 formalization of Fricke material. The Fermat’s Last Theorem artifact [10] is a Lean 4 development built on Mathlib that already contains substantial Fricke and Atkin–Lehner theory. A repository-scoped code search returns on the order of 2.7×10^3 Lean files mentioning Fricke and 3.6×10^3 mentioning AtkinLehner, including `Thm_ModularForm_heckeU_add_slash_fricke_eq_zero`, `Thm_CuspForm_hasNebentypus_inv_and_qCoeff_hecke_eigen_of_fricke` and `Def_ModularCurve_AtkinLehner`. Their statements use the same matrix $\begin{pmatrix} 0 & -1 \\ N & 0 \end{pmatrix}$ and the same Mathlib slash action that we use here, and they go considerably further — to Hecke operators, eigenforms with nebentypus, and Atkin–Lehner operators on modular curves.

Most directly, `Def_CuspForm_AtkinLehnerOperator.lean` defines `atkinLehnerLin`, a map `ModularForm (Γ0(M)) k →ℓ[C] ModularForm (Γ0(M)) k`, discharging slash-invariance, holomorphy and boundedness at the cusps as our `frickeModularOperator` does — our (C5) and (C6), for the whole Atkin–Lehner family and as a $\mathbb{C}$ -linear map. Their `Def_ModularForm_AtkinLehnerDatum.lean` builds its group element with the same `mkOfDetNeZero` idiom we use and proves the analogue of our (C1), (C2) and (C7).

In particular their `CohCarrier.frickeMat` is, up to naming, our `frickeConj`: the same conjugate matrix $\begin{pmatrix} d & -c/N \\ -Nb & a \end{pmatrix}$, the same divisibility hypothesis, and a determinant proof that likewise finishes with `linear_combination`. Our development was arrived at independently, which does not bear on priority; we record the coincidence because a reader comparing the two should know they are the same construction.

What is nonetheless offered here. The FLT material is proof-internal: it lives in a bespoke `CohCarrier` namespace, and W_N is generally passed to each theorem as a parameter constrained by a matrix equation. We instead define W_N once as a `GL (Fin 2) ℝ` element and prove the small surrounding theory — membership in GL_2^+ , $W_N^2 = -N \cdot I$, the conjugation formula, and the induced operators on `SlashInvariantForm` and `ModularForm`. The intended destination is Mathlib, which has none of this. Stated precisely, what is left after disclosure is: the literal

matrix W_N over $\mathbb{R}$ (their datum’s lower-right entry is q , so it does not specialize to W_N on the nose), the explicit composite scalar of (C7), and a four-named-import development using stock tactics rather than a bespoke proof harness. That is a methodology difference, and whether it is worth anything is a question for Mathlib review, not for us to assert.

7 Limitations

Fricke and Atkin–Lehner operators are classical; the definitive treatment of the Atkin–Lehner involutions W_Q for $Q||N$ is [1], and textbook accounts appear in [4, 5, 6]. Our contribution is not mathematical novelty but the transfer of this material into a machine-checked setting where it can be depended upon.

We state the limitations plainly.

Remark 7.1 (What is *not* done here).

- (a) We formalize the Fricke involution W_N only. The full Atkin–Lehner family W_Q for $Q||N$, and the resulting simultaneous eigenspace decomposition, are not treated.
- (b) The ± 1 eigenspace decomposition of $M_k(\Gamma_0(N))$ is not itself packaged as a Lean definition. The mathematical content is Theorem 4.3, which is formalized; what remains is bookkeeping — carrying the even- k hypothesis through the `zpow` arithmetic on the constant and bundling the two eigenspaces as submodules.
- (c) We make no priority claim. Our gap check was run against Mathlib, where it holds; a later search of the Lean 4 developments cited here found the prior art described in §6. We have not searched the Lean Zulip or other proof assistants.

8 Intended application

The motivation is downstream. Eta-quotients $f(\tau) = \prod_{\delta|N} \eta(\delta\tau)^{r_\delta}$ are modular on $\Gamma_0(N)$ under Ligozat’s conditions, and their behaviour under Fricke and Atkin–Lehner operators is what makes multiplicativity tractable [7]. A formalization of eta-quotient identities therefore needs W_N as a prerequisite, which is what this module supplies. A first attempt to make that link precise is reported in §9 below, and it failed instructively rather than succeeding. With the operator now defined on `ModularForm`, the immediate next steps are the normalization constant of limitation (b) and the full Atkin–Lehner family of limitation (a). On the analytic side, the Rademacher-type coefficient bounds that accompany the eta-quotient programme will first be checked at the program-verified tier with lightweight inequality-proof tooling in the style of Tao’s *estimates* assistant [14], and only then promoted to kernel-checked statements — the same two-tier discipline (exact checker below, kernel above) used throughout this programme.

9 A negative result: an attempted T-duality bridge

A follow-up effort, reported here for the record rather than left as an unqualified forward pointer, attempted to formalize that W_N is not merely a device for eta-quotients but the Lean 4 incarnation of a physical duality: that T-duality’s action on the complex-structure modulus τ of a T^2 factor, restricted to $\Gamma_0(N)$, *coincides* with W_N ’s already-proven action on $\mathbb{H}$. This section states plainly what was found, because a withheld negative result is exactly the failure mode this programme has already corrected twice before [10] — and because what survives the attempt is genuine and reusable.

What is proved, sorry-free, and reusable

Theorem 9.1 (W_N 's Möbius action, in closed form). *frickeW_smul_coe*: for every $N > 0$ and $\tau \in \mathbb{H}$,

$$(W_N \cdot \tau : \mathbb{C}) = \frac{-1}{N\tau}.$$

This closes a real seam that nothing in the library previously closed: the $-1/(N\tau)$ normal form is exactly what the $\mathbb{C}$ -valued eta-quotient half of this programme already computes with (`eta_fricke_factor`, in the eta-quotient modules), while W_N itself had only ever been stated in $GL_2(\mathbb{R})$ -action language. The proof goes through Mathlib's determinant case split (Theorem 3.1 supplies exactly the positive-determinant hypothesis Mathlib's own action lemma needs, so no hidden sign or conjugation enters) and is certified non-vacuous by a negative control: the identical proof script against four independently-computed *wrong* right-hand sides ($+1/(N\tau)$, $-1/\tau$, $-N/\tau$, $-1/(N + \tau)$) fails with four distinct **unsolved goals**, each displaying the same correct left-hand side.

What is not proved, and why

The bridge statement itself,

$$(W_N \cdot \tau : \mathbb{C}) = \frac{-1}{N\tau} \wedge \forall \gamma \in \Gamma_0(N), \gamma \cdot \tau \text{ is the standard fractional-linear action of } SL_2(\mathbb{Z}),$$

remains **sorry**, under a build-failing inverted axiom guard, and an adversarial review of this programme's own workflow found two independent reasons it should stay that way.

It is a relabelling, not a reduction. A standalone equivalence check, compiled and verified, shows the bridge statement is logically equivalent to the conjunction of three already-proved lemmas and adds no content beyond them: the hypothesis $\gamma \in \Gamma_0(N)$ is never consumed (the second conjunct holds for every $\gamma \in SL_2(\mathbb{Z})$), and neither Theorem 3.4 (that W_N normalizes $\Gamma_0(N)$) nor the involution property $W_N^2 = -N \cdot I$ occurs anywhere in the statement or its proof term. Nothing in the Lean term names a duality group, a string background, or a spectrum; the physics is asserted only in prose around the theorem, not inside it.

The physics citation does not support it. The natural reference for a τ -modulus T-duality action, Giveon–Porrati–Rabinovici [8], anchors $\tau \mapsto -1/\tau$ only at $N = 1$. For $N > 1$, $\det W_N = N$, so W_N is not even an element of the $SL_2(\mathbb{Z})$ that reference describes, and an exhaustive search of its text returns no occurrence of a Fricke involution, a congruence subgroup, or a level-dependent duality element at any $N > 1$. The correct physical home for a level- N Fricke-type map turns out to be a different object entirely: Persson–Volpato [9] identify $S \mapsto -1/(NS)$ as an *S-duality* on the heterotic axio-dilaton modulus in CHL orbifold models, explicitly *outside* the standard $SL_2(\mathbb{Z})$ symmetry of the parent theory — a different modulus, and a different duality, than the one this attempt targeted.

The honest conclusion

Theorem 9.1 is real: it is evidence that W_N is the right mathematical object for a physical correspondence, stated purely as a fact about $\mathbb{H}$ and $\mathbb{C}$ with no physics content claimed. Which correspondence, and under what hypotheses, remains open, and the Persson–Volpato reference is where the next attempt should start — inside CHL orbifolds of $K3 \times T^2$, on the axio-dilaton, not the T^2 modulus. We record this as a finding rather than quietly dropping the sentence that used to promise it.

10 Conclusion

We have formalized the Fricke involution W_N on $\Gamma_0(N)$ in Lean 4 and the operator it induces on modular forms, addressing a gap in *Mathlib* — though not, as §6 records, in Lean 4 as a

whole. The development is small — 27 declarations in 291 lines — depends only on Lean’s three standard axioms, and is built on Mathlib’s existing `SlashAction`, `SlashInvariantForm` and `ModularForm` infrastructure rather than duplicating it, so that it composes with the rest of the library. The mathematics is classical; the machine-checked statement of it is not.

A note on method: AI assistance and human responsibility

This note follows the disclosure norm argued for by Tao [12], endorsing recommendations of the Leiden Declaration on Artificial Intelligence and Mathematics [13]: disclose tool use, affirm that credit and responsibility belong to the human author, and attribute sources where AI tools cannot reliably do so themselves.

What AI did. The Lean 4 formalization, the literature search, and the drafting of this text were produced with the assistance of large language models (Anthropic’s Claude family), working under human direction across many sessions. The methodology is neuro-symbolic in the specific sense Tao describes [12]: natural-language reasoning proposes definitions, proof strategies, and prose, and the Lean kernel — not the model, and not the author’s trust in the model — is what admits or rejects every mathematical claim. Every theorem in this paper compiles against Mathlib with no `sorry` and a build-failing axiom guard on its footprint (§5); nothing here is asserted true because a model said so.

What the author did. Xavier Callens set the research direction, chose what to formalize and in what order, judged what the results mean and whether they are worth publishing, and is responsible for every claim in this document, including its errors. That responsibility was exercised concretely, not nominally, in this paper’s own history: an earlier draft claimed priority for this formalization, and that claim was found false, retracted, and replaced with the attribution recorded in §6; a follow-up attempt to extend W_N to a physical duality (§9) was found, on adversarial review, to reduce to a relabelling with an unsupported physics citation, and is reported here as a negative result rather than closed as a theorem. Tao’s own test is apt: a result that cannot be explained and defended by its author, formally verified or not, is not ready to be called complete [12]. The mathematical content of W_N — what it is, why it normalizes $\Gamma_0(N)$, and why the two withdrawn claims above were in fact wrong — is something the author can explain without further AI assistance; the Lean proof terms that certify it were substantially AI-drafted and are trusted only because the kernel, not the author, checks them.

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